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Adaptive learning from noisy estimated depth maps benefits monocular RGB-based 3D human pose estimation

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ABSTRACT

Monocular RGB-based 3D human pose estimation has broad applications in human-centric scenarios. Mainstream methods follow a 2D-to-3D lifting pipeline, that first infers a 2D pose from RGB images then lifts it to 3D. However, this pipeline still struggles with depth ambiguity, despite recent advances in implicitly inferring depth cues from RGB images or temporal information from 2D pose sequences. Inspired by recent advances in dedicated depth estimation, this paper aims to explicitly incorporate depth cues through estimated depth maps to reduce depth ambiguity. Nevertheless, such estimated depth maps often suffer from substantial irrelevant background and noise. To address these two issues, we propose the Adaptive Sample and Fusion Network (ASFnet). First, we present an Adaptive Multi-Sampling (AMS) module that utilizes estimated 2D poses as guidance to focus on pose-relevant depth information. To reduce the influence of 2D pose noise, AMS adaptively adjusts sampling points and weights using both the depth map and 2D pose, focusing on informative regions. Second, we propose an Uncertainty-aware Depth Enhancer (UDE) module to adaptively fuse depth features by suppressing depth map noise. Guided by a heteroscedastic uncertainty loss, UDE estimates the mean and variance of depth features for each joint, with variance indicating uncertainty to filter unreliable information. Subsequently, the adjusted depth features and 2D poses are integrated via a Multimodal Interaction Transformer using cross-modal attention. Extensive experiments on Human3.6M and MPI-INF-3DHP verify our method's effectiveness. Our code is available at https://github.com/liujingtingdev/ASFnet.

1. Introduction

Monocular RGB-based 3D human pose estimation aims to infer 3D poses from single-view images, playing a vital role in human-centric applications such as content generation [1,2] and action recognition [3,4]. With the advancement of robust 2D pose detectors [5] and the availability of large-scale 3D motion capture datasets [6], the dominant 2D-to-3D lifting pipeline first infers 2D poses from RGB images and then lifts them to 3D. However, this pipeline remains fundamentally ill-posed due to depth ambiguity-multiple 3D poses can project to identical 2D observations. While existing methods have made progress, those relying solely on 2D keypoints [7,8] or spatial context features [9] fail to resolve this ambiguity. Even methods that employ explicit depth modeling [10,11] are constrained by coarse ordinal relations and simply fuse the estimated depth cues with 2D poses, which cannot provide the precise depth information required for 3D pose estimation.

A promising direction is to incorporate a dedicated depth estimator to extract depth maps, providing direct and precise depth constraints to alleviate ambiguity. However, this approach introduces two critical challenges. First, to extract pose-relevant information from the whole depth map, a straightforward strategy is to sample the depth map at estimated 2D joint locations. Inaccuracies in the 2D pose can lead to misaligned sampling, propagating erroneous depth features to the 3D pose. Second, the depth maps themselves inevitably contain noise due to estimation errors, especially under challenging conditions like motion blur or occlusion. To address these challenges, we propose a novel depth-aware framework whose core novelty lies in the synergistic combination of two dedicated modules, each designed to tackle a distinct fundamental weakness. As illustrated in Fig. 1 (b), our solution directly maps a dedicated module to each challenge: the Adaptive Multi-Sampling (AMS) module tackles spatial misalignment by dynamically adjusting sampling points around estimated 2D joints to extract reliable pose-relevant depth features. Concurrently, the Uncertainty-aware Depth Enhancer (UDE) module suppresses depth noise by explicitly modeling the uncertainty of each joint's depth features, self-adaptively emphasizing reliable cues and suppressing noisy ones. By integrating these two

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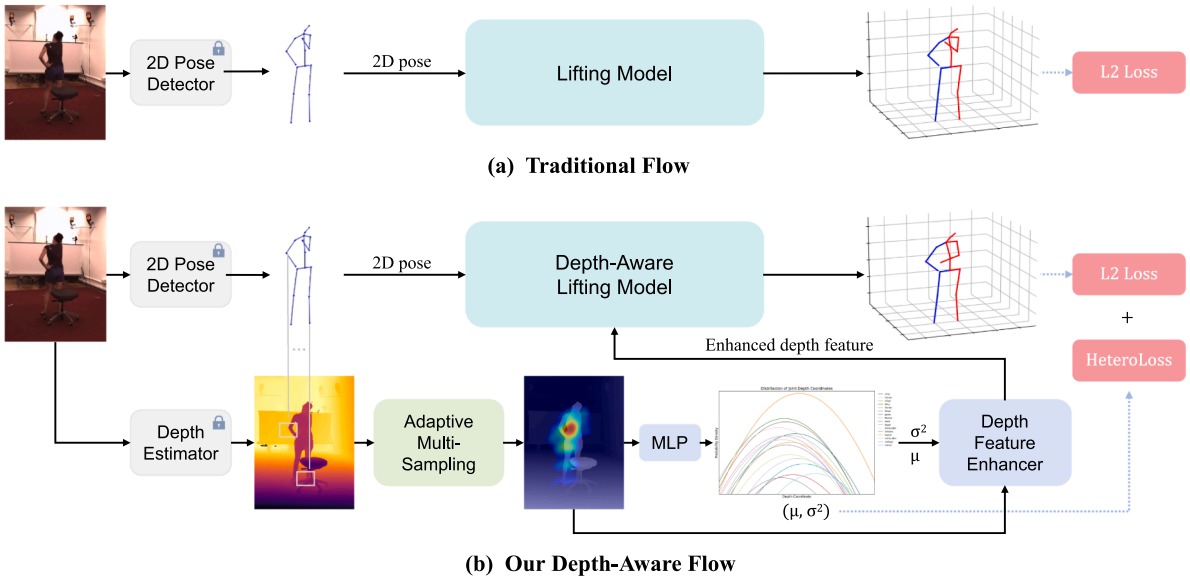


Fig. 1. Comparison between traditional lifting methods and our depth-aware approach for monocular RGB-based 3D human pose estimation. Traditional lifting methods (a) estimate 3D pose by lifting 2D poses extracted from RGB images, relying solely on planar information and suffering from depth ambiguity. Our method (b) incorporates a dedicated depth estimator to capture explicit depth cues from the RGB image, mitigating this ambiguity. We propose Adaptive Multi-Sampling to dynamically focus on pose-relevant regions, improving feature extraction despite noisy 2D poses. The Uncertainty-aware Depth Enhancer further refines depth features by suppressing noise. Finally, fused depth and 2D features enable our model to regress more accurate 3D pose.

modules, our framework establishes a robust pipeline that directly addresses the two distinct weaknesses in depth-aware pose estimation, transforming raw depth maps into informative and reliable cues.

Specifically, to address the challenge of extracting reliable pose-relevant depth features in the presence of noisy 2D poses, AMS adaptively learns a set of offsets and weights from both the depth map and the estimated 2D pose, and uses them to perform flexible multi-sampling from the depth map. This design allows dynamic adjustment of sampling points during training. By doing so, AMS enables the model to capture rich pose-relevant depth information while mitigating the impact of errors in the 2D pose estimation.

Furthermore, to suppress inherent depth noise and improve robustness under varying qualities of depth estimation. UDE predicts a distribution over the sampled depth features for each joint by estimating the expected value as the depth reference and the variance as the corresponding uncertainty. This uncertainty is then utilized to enhance the depth features by emphasizing reliable information and suppressing noise. To achieve this, we formulate the refinement process as a heteroscedastic uncertainty learning task, where the depth distribution of each joint is constrained through a probabilistic loss function. The resulting variance matrix explicitly quantifies the measurement uncertainty, allowing the model to self-adaptively boost informative depth cues. By incorporating this uncertainty-aware refinement, UDE significantly improves the stability and reliability of depth features in the face of estimation errors.

Finally, a Transformer-based fusion regressor integrates the refined multi-modal features for 3D pose estimation. Notably, the proposed AMS and UDE modules are highly modular and flexible, and can be readily integrated into existing lifting-based frameworks, including both single-frame and multi-frame variants. The efficacy of our depth-aware approach is demonstrated in our experiments. Our framework not only sets a new state-of-the-art on benchmark datasets, outperforming existing lifting-based and fusion methods, but also delivers a critical practical advantage: it exhibits remarkable robustness, significantly mitigating the performance penalty typically associated with using efficient, low-precision depth estimators. These gains quantitatively validate that directly tackling the dual problems of spatial misalignment and depth noise is key to advancing depth-aware 3D pose estimation. The main contributions of this work are summarized as follows:

- A novel framework that synergistically combines adaptive sampling with uncertainty-aware enhancement to address the core challenges in depth-aware 3D pose estimation: spatial misalignment from imperfect 2D poses and inherent noise in depth maps.
- We introduce an Adaptive Multi-Sampling (AMS) module that progressively refines sampling locations through iterative correction to extract comprehensive pose-relevant features.
- We introduce an Uncertainty-aware Depth Enhancement (UDE) module, which suppresses noise and enhances feature reliability by modeling depth feature distributions with quantified uncertainty.
- Experiments on benchmark datasets for 3D human pose estimation, i.e., Human3.6M [12] and MPI-INF-3DHP [13], demonstrate that ours achieves competitive performance with state-of-the-art methods.

2. Related work

Recent deep learning methods for monocular 3D human pose estimation have advanced significantly. Early approaches followed two distinct paths: some [14] utilized depth cameras to estimate 3D poses from depth images, but their applicability was limited by hardware requirements; others [15] directly regressed 3D poses from RGB images, yet were constrained by available 3D motion capture data. With progress in 2D pose estimation [5,16], the lifting paradigm-first estimating 2D poses then converting to 3D-has emerged as a prominent approach, simplifying the estimation pipeline.

2.1. *Lifting-based methods*

Lifting-based approaches simplify 3D pose estimation by leveraging off-the-shelf 2D pose detectors trained on diverse datasets and lifting their outputs to 3D, a paradigm pioneered by [6]. To address depth ambiguity, methods model spatio-temporal dependencies using LSTMs [17] and CNNs [18]. Several works [8] further incorporate human skeletal constraints via graph neural networks. Recently, transformer-based methods [7] have gained significant attention, building on their success in vision tasks. The latest advancement employs diffusion models, which learn valid 3D pose distributions to synthesize diverse samples. This powerful generative paradigm addresses data scarcity and

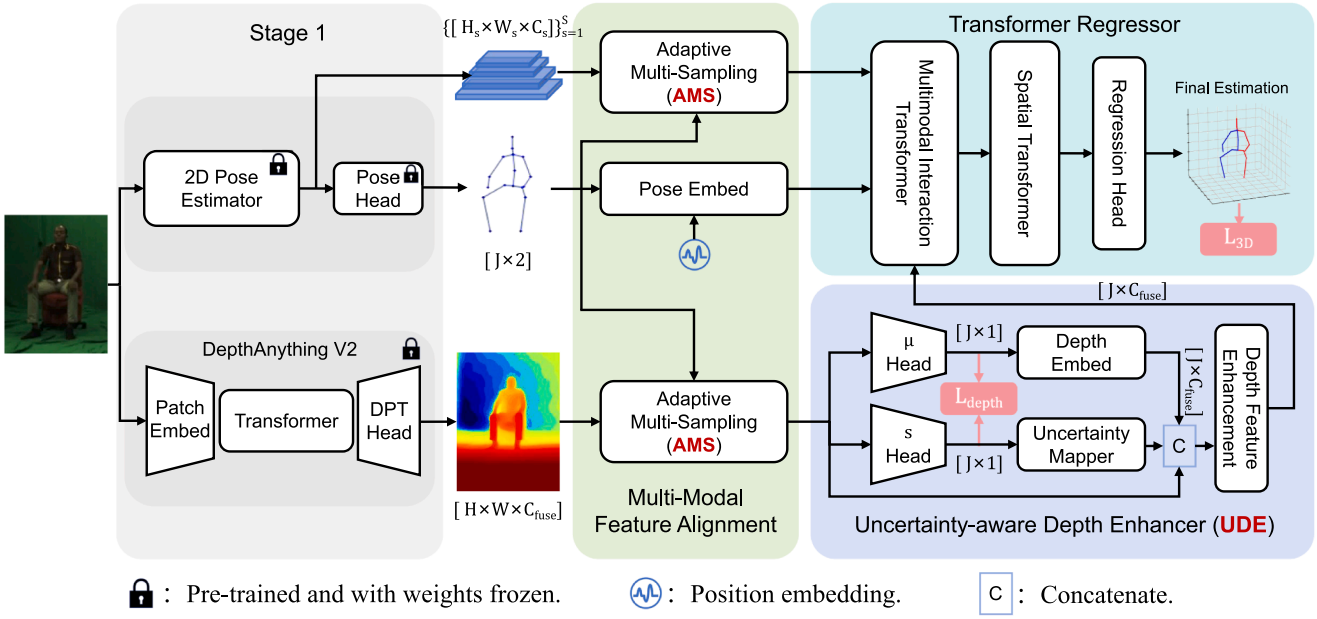


Fig. 2. Overview of our Adaptive Sample and Fusion Network (ASFnet). The process begins by obtaining the 2D pose from a pre-trained 2D Pose Estimator and the depth map from a pre-trained Depth Estimator. Guided by the 2D pose, the Adaptive Multi-Sampling (AMS) module samples both image features and depth features from the RGB image features and the depth map. Since depth features often contain noise, the Uncertainty-aware Depth Enhancer (UDE) is applied to further refine and enhance them. Subsequently, the Multimodal Interaction Transformer fuses the embedded 2D pose features, image features, and the enhanced depth features. The combined features are then fed into a regressor to predict the final 3D human pose. Note that the Multimodal Interaction Transformer is implemented using a vanilla Transformer architecture, and the depth estimator employed is the state-of-the-art DepthAnything V2.

poor generalization by serving as probabilistic data augmentors, enriching the training hypothesis space and enhancing robustness, particularly in low-data regimes. For example, some works [19–21] condition the reverse process on 2D keypoints for geometric consistency, while others [22,23] embed anatomical priors for biomechanical feasibility. The framework has also been extended for zero-shot cross-domain generalization [24] and robust multi-frame video estimation [25].

Despite advances, methods relying solely on 2D poses face the ineluctable challenge of depth ambiguity. Our work directly addresses this by incorporating explicit robust depth features to guide the lifting model.

2.2. Fusion methods

Beyond mainstream approaches, some methods integrate 2D poses with image information to compensate for insufficient 2D cues. However, these methods face several limitations: some bypass established 2D detectors [26], increasing complexity while losing detector benefits; others employ suboptimal fusion strategies, such as segmenting local patches [27] or using data-sparse heatmaps [21]. While CA-PF [9] queries global features with 2D poses and PSTR [28] improves training, these methods still struggle with the inherent inverse problem and lack explicit depth cues from 2D-derived features. Other approaches explicitly target depth estimation: some design specialized modules for joint depth ranking [10], while Nie et al. [11] regresses depth from local patches via LSTM. Yet these methods suffer from limited accuracy due to simplistic fusion process and local processing. Alternatively, methods relying on depth camera datasets [29] face cost and generalization barriers, restricting their applicability to general RGB imagery.

In contrast to prior works, our method delivers reliable depth cues for pose estimation without relying on depth cameras, achieved through an uncertainty-aware depth enhancement mechanism.

3. Method

3.1. Overview

Fig. 2 illustrates our overall framework, with key notations summarized in Table 1. Given RGB images $\mathbf{I} \in \mathbb{R}^{B\times H\times W\times 3}$, we employ a pre-trained 2D pose detector to extract: 2D poses $\mathbf{p} \in \mathbb{R}^{B\times J\times 2}$ and multi-scale image features $\{\mathbf{H}s \in \mathbb{R}^{B\times H_s\times W_s\times C_s}\}_{s=1}^{S}$. A depth estimator extracts depth features $\mathbf{D} \in \mathbb{R}^{B\times H\times W\times C_{\text{fuse}}}$. The Multi-Modal Feature Alignment module then aligns depth and image features with the 2D pose via Adaptive Multi-Sampling (AMS), extracting pose-related features while suppressing background. To mitigate depth noise, UDE enhances features by downweighting unreliable regions using confidence scores. Finally, the Transformer Regressor fuses all modalities via the Multimodal Interaction Transformer for final regression.

3.2. Adaptive multi-sampling (AMS)

The estimated depth and image feature maps encompass substantial background information, introducing both interference and computational overhead. To address this, we align these maps with the 2D pose to extract pose-relevant information while suppressing background noise.

We employ grid sampling guided by 2D poses for initial alignment. However, sole reliance on estimated 2D poses is suboptimal due to detector inaccuracies. We therefore propose the Adaptive Multi-Sampling (AMS) module, which iteratively refines sampling locations through multiple sampling stages to correct initial pose errors, thereby enhancing feature quality and the capture of complex joint spatial relationships.

Pose Embed. Given estimated 2D poses $\mathbf{P} \in \mathbb{R}^{B\times J\times 2}$, the Pose Embed module first transforms joint coordinates (x, y) via linear projection to high-dimensional features, augments them with positional encoding, and applies dropout for regularization, producing embedded pose features $\mathbf{P}' \in \mathbb{R}^{B\times J\times C_{fuse}}$.

Adaptive Multi-Sampling. The structure of AMS is shown in Fig. 3. We use the estimated 2D pose and features to adaptively learn multiple sets of offsets $\mathbf{\Delta} \in \mathbb{R}^{B\times J\times N_2\times N_1\times 2}$ and corresponding weights

Table 1
Key mathematical notations used in our methodology.

Notation	Description
J	Number of human joints.
N_1	Number of sampling offsets (or weights) per head in the AMS module.
N_2	Number of heads in the AMS module.
C_{fuse}	The channel dimension of feature representations.
S	Number of multi-scale image feature sets extracted from the 2D pose detector.
H, W	Height and width of the images.
$\mathbf{P} \in \mathbb{R}^{B\times J\times 2}$	The initial estimated 2D pose (pixel coordinates).
$\mathbf{P}' \in \mathbb{R}^{B\times J\times C_{\text{fuse}}}$	The embedded 2D pose features.
$\mathbf{D} \in \mathbb{R}^{B\times H\times W\times C_{\text{fuse}}}$	The feature representation of the input depth map.
$\mathbf{D}' \in \mathbb{R}^{B\times J\times C_{\text{fuse}}}$	The initially sampled depth features at $\mathbf{P}$.
$\mathbf{\Delta} \in \mathbb{R}^{B\times J\times N_2\times N_1\times 2}$	The learned sampling offsets in AMS.
$\mathbf{W} \in \mathbb{R}^{B\times J\times N_2\times N_1}$	The learned attention weights for the sampled points in AMS.
θ_o, θ_w	The offset and weight prediction heads (MLPs) in AMS.
$\mathbf{F}_d \in \mathbb{R}^{B\times J\times C_{\text{fuse}}}$	The final, refined depth features output by the AMS module.
$\{\mathbf{F}_s\}_{s=1}^{S}$	The set of S image feature sets, where each $\mathbf{F}_s \in \mathbb{R}^{B\times J\times C_{\text{fuse}}}$ is obtained by applying the AMS module to image features from the 2D detector.
$\mu_j \in \mathbb{R}^{B\times 1}$	The coarse estimated depth value for joint j. The vector $\boldsymbol{\mu} \in \mathbb{R}^{B\times J\times 1}$ contains depth estimates for all joints.
$s_j \in \mathbb{R}^{B\times 1}$	The log-variance for joint j in the UDE module. The vector $s \in \mathbb{R}^{B\times J\times 1}$ contains log-variances for all joints.
$\text{Head}_\mu, \text{Head}_s$	MLP heads for predicting μ_j and s_j in UDE.
$\mathbf{F}_d^{cat} \in \mathbb{R}^{B\times J\times (C_{\text{fuse}}\times 3)}$	The concatenation of $\mathbf{F}_d$, $\text{Head}_\mu(\mu)$ and $\text{Head}_s(s)$, serving as the input to the Depth Feature Enhancement in UDE.
$\widetilde{\mathbf{F}}_d \in \mathbb{R}^{B\times J\times C_{fuse}}$	The enhanced depth features output by the UDE module.
$\mathbf{F} \in \mathbb{R}^{B\times J\times (S+2)\times C_{fuse}}$	The stacked multi-modal features $(\mathbf{P}', \{\mathbf{F}_s\}_{s=1}^{S}, \widetilde{\mathbf{F}}_d)$ for final pose regression.

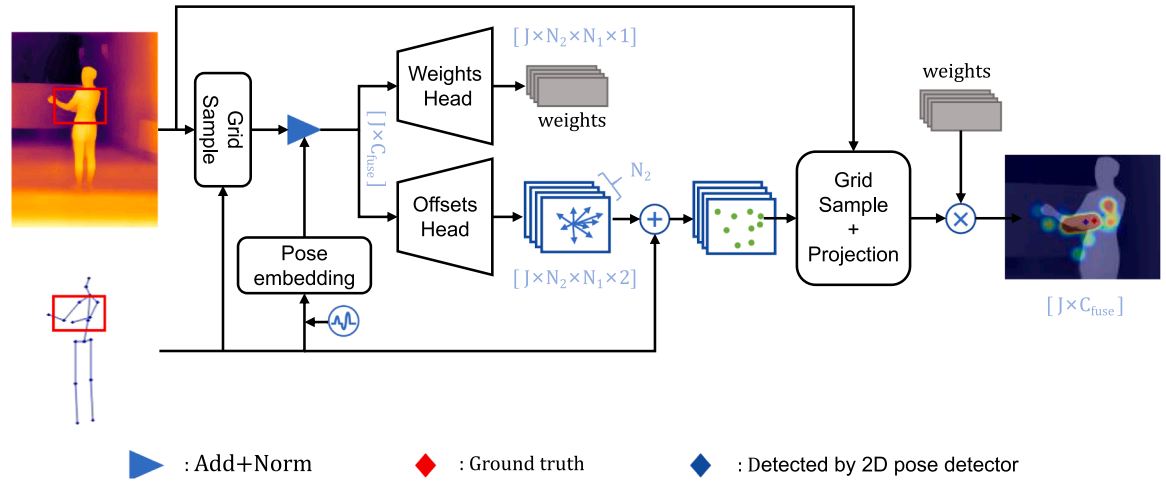


Fig. 3. Overview of our Adaptive Multi-Sampling (AMS) module, which adaptively learns offsets and weights to generate multiple sets of sampling points with varying importance, enabling it to capture more pose-related information. AMS overcomes the limitation of relying solely on the estimated 2D pose, which is constrained by the accuracy of the 2D pose estimator. For better visualization, we apply Gaussian blur to convert the sampled joints into heatmaps, where higher weights correspond to brighter colors.

$\mathbf{W} \in \mathbb{R}^{B\times J\times N_2\times N_1}$, where 2 represents the x and y coordinates and N_1 denotes the number of offsets or weights in one head. There are a total of N_2 heads, and each head adaptively learns distinct sampling points from depth maps, enabling features to capture both joint locations and surrounding context for richer pose information. The offsets are then added to the initial 2D pose to obtain sampling points, with the process formulated as:

$$\mathbf{H} = \text{Norm}(\mathbf{D}' + \mathbf{P}'), \tag{1}$$

$$\mathbf{\Delta} = \theta_o(\mathbf{H}), \tag{2}$$

$$\mathbf{W} = \theta_w(\mathbf{H}), \tag{3}$$

where θ_o and θ_w denote the Offsets Head and Weights Head respectively. Both operations are implemented as fully connected layers. Specifically, the Weights Head generates attention weights, which are then normalized by softmax. The Offsets Head produces sampling offsets, which are subsequently transformed by a tanh activation. $\mathbf{D}'$ denotes the initial sampled depth features, obtained by sampling from the initial 2D pose. To align the dimensions with the 2D pose, an initial sampling operation is performed on $\mathbf{D}$, ensuring that the features are consistent with the pose's spatial representation.

Then, these offset sampling points are used to sample the input feature maps, and finally, the sampled features are weighted according to the learned weights, resulting in the depth features $\mathbf{F}_d \in \mathbb{R}^{B\times J\times C_{fuse}}$. The process is formulated as:

$$\mathbf{D}_{sample} = \text{GridSample}(\mathbf{D}, \mathbf{\Delta} + \mathbf{P}), \tag{4}$$

$$\mathbf{F}_d = \text{Projection}(\mathbf{D}_{sample}) \cdot \mathbf{W}, \tag{5}$$

The GridSample operation samples the feature map within the normalized coordinate range of $[-1, 1]$, with coordinates outside this range yielding zero-valued features. The Projection$(\cdot)$ refers to a linear layer that maps the sparsely sampled points to the feature space.

The above formulation is illustrated using depth maps as an example. Similarly, we perform the same operations on image features that have been extracted by the 2D detector, yielding the feature set $\{\mathbf{F}_s \in \mathbb{R}^{B\times J\times C_{fuse}}\}_{s=1}^{S}$.

Forward Pass. The Adaptive Multi-Sampling (AMS) module structure is shown in Fig. 3 and implemented in Algorithm 1. Taking the embedded pose $\mathbf{P}'$ and initial sampled depth features $\mathbf{D}'$ (both in $\mathbb{R}^{B\times J\times C_{\text{fuse}}}$) as input, AMS first fuses them via addition and normalization. From this unified representation, it predicts sampling configurations across N_2 independent heads, with each head yielding N_1 outputs, generating offset vectors $\mathbf{\Delta}\in\mathbb{R}^{B\times J\times N_2\times N_1\times 2}$ and corresponding attention weights $\mathbf{W}\in\mathbb{R}^{B\times J\times N_2\times N_1}$. Sampling locations are computed by adding $\mathbf{\Delta}$ to the original 2D pose coordinates $\mathbf{P}$. A grid sampler subsequently samples the depth feature map $\mathbf{D}$ at these $N_2\times N_1$ points per joint. The sampled features are then projected to a dimension of $\mathbb{R}^{B\times J\times N_2\times N_1\times(C_{\text{fuse}}/N_2)}$ and aggregated via a weighted sum over the N_1 dimension using the attention weights $\mathbf{W}$, ultimately producing the refined output $\mathbf{F}_d\in\mathbb{R}^{B\times J\times C_{\text{fuse}}}$. This AMS module is stacked four times to enable progressive feature refinement.

Algorithm 1 Forward pass of adaptive multi-sampling (AMS) module.

Input: The estimated 2D pose $\mathbf{P}$, depth feature map $\mathbf{D}$, and corresponding Embedded pose $\mathbf{P}'\in\mathbb{R}^{B\times J\times C_{\text{fuse}}}$, initial sampled depth features $\mathbf{D}'\in\mathbb{R}^{B\times J\times C_{\text{fuse}}}$
Output: Refined depth features $\mathbf{F}_d\in\mathbb{R}^{B\times J\times C_{\text{fuse}}}$

$\mathbf{F}_d\leftarrow\mathbf{D}'$;
for $t=1$ **to** *iterations* **do**
$\mathbf{F}_{\text{fuse}}\leftarrow\text{AddNorm}(\mathbf{P}'+\mathbf{F}_d)$;
$\mathbf{\Delta}\leftarrow\theta_o(\mathbf{F}_{\text{fuse}})$ //$\mathbf{\Delta}\in\mathbb{R}^{B\times J\times N_2\times N_1\times 2}$
$\mathbf{W}\leftarrow\theta_w(\mathbf{F}_{\text{fuse}})$ //$\mathbf{W}\in\mathbb{R}^{B\times J\times N_2\times N_1}$
$\mathbf{L}\leftarrow\mathbf{P}+\mathbf{\Delta}$ //Compute sampling locations
$\mathbf{D}_{\text{sample}}\leftarrow\text{GridSample}(\mathbf{D},\mathbf{L})$ //Sample features
$\mathbf{D}'_{\text{sample}}\leftarrow\text{Projection}(\mathbf{D}_{\text{sample}})$ //Project to $\mathbb{R}^{B\times J\times N_2\times N_1\times(C_{\text{fuse}}/N_2)}$
$\mathbf{F}_d\leftarrow\sum_{n_1=1}^{N_1}\mathbf{W}[\ldots,n_1]\odot\mathbf{D}'_{\text{sample}}[\ldots,n_1]$;
$\mathbf{F}_d\leftarrow\text{ConcatHeads}(\mathbf{F}_d)$;
end

3.3. Uncertainty-aware depth enhancer (UDE)

Depth maps obtained from monocular depth estimation models often suffer from inherent noise due to limited precision. Directly fusing the sampled depth features with other modalities (e.g., RGB features or 2D pose) may propagate unreliable depth information. To address this challenge, we propose an Uncertainty-aware Depth Enhancer (UDE) that dynamically modulates the depth features before fusion based on their reliability. Inspired by uncertainty modeling [30] in computer vision, we explicitly model the uncertainty of depth features at each joint to evaluate their reliability (higher uncertainty indicates noisier features), making UDE is able to adaptively optimize the depth features based on uncertainty, thereby improving stability against depth map imperfections.

We formulate depth feature enhancement as a probabilistic learning problem. For each joint j, we assume its depth feature follows a Gaussian distribution $\mathcal{N}(\mu_j,\sigma_j^2)$, where μ_j represents the estimated depth value, which relies solely on depth features $\mathbf{F}_d^{(j)}$, and σ_j^2 quantifies its uncertainty. To ensure numerical stability, the network regresses the log-variance $s_j=\log\sigma_j^2$ instead of directly predicting σ_j^2:

$$\mu_j=\text{Head}_\mu(\mathbf{F}_d^{(j)}),\tag{6}$$

$$s_j=\text{Head}_s(\mathbf{F}_d^{(j)}),\tag{7}$$

where Head_μ and Head_s share identical architectures, each comprising a MLP with GeLU activations. $\mu,s\in\mathbb{R}^{B\times J\times 1}$.

To enable dynamic enhancement of depth features $\mathbf{F}_d$ by coarse depth estimates μ and uncertainty scores s, we first project these scalars into learnable high-dimensional embeddings: $\mathbf{F}_d^{cat}=\text{Concat}(\mathbf{F}_d,\text{DEmbed}(\mu),\text{UMap}(s))$. Here, $\text{DEmbed}(\cdot)$ denotes the Depth Embed module—a linear projection layer with positional encoding that maps inputs to spatially-aware feature embeddings. $\text{UMap}(\cdot)$ refers to the Uncertainty Mapper—a linear projection layer with softmax activation that generates uncertainty score maps.

The Depth Feature Enhancement module then adaptively refines $\mathbf{F}_d^{cat}$ through multi-head self-attention (MSA) and MLP projection. During refinement, uncertainty scores and coarse depth values guide the network to amplify reliable depth cues while suppressing noisy regions via uncertainty-aware modulation. Implementation proceeds as follows:

$$\mathbf{F}_d^{attn}=\text{MSA}(\mathbf{F}_d^{cat})+\mathbf{F}_d^{cat},\tag{8}$$

$$\widetilde{\mathbf{F}}_d=\text{MLP}(\mathbf{F}_d^{attn})+\mathbf{F}_d^{attn}.\tag{9}$$

Algorithm 2 Forward pass of uncertainty-aware depth enhancer (UDE) module.

Input: Refined depth features $\mathbf{F}_d\in\mathbb{R}^{B\times J\times C_{\text{fuse}}}$
Output: Enhanced depth features $\widetilde{\mathbf{F}}_d\in\mathbb{R}^{B\times J\times C_{\text{fuse}}}$

$\boldsymbol{\mu}\leftarrow\text{Head}_\mu(\mathbf{F}_d)$ //$\boldsymbol{\mu}\in\mathbb{R}^{B\times J\times 1}$
$\boldsymbol{s}\leftarrow\text{Head}_s(\mathbf{F}_d)$ //$s\in\mathbb{R}^{B\times J\times 1}$
$\mathbf{F}_d^{\text{cat}}\leftarrow[\mathbf{F}_d;\text{DEmbed}(\boldsymbol{\mu});\text{UMap}(\boldsymbol{s})]$;
$\mathbf{F}_d^{\text{attn}}\leftarrow\text{MSA}(\mathbf{F}_d^{\text{cat}})+\mathbf{F}_d^{\text{cat}}$ //Multi-head self-attention with residual
$\widetilde{\mathbf{F}}_d\leftarrow\text{MLP}(\mathbf{F}_d^{\text{attn}})+\mathbf{F}_d^{\text{attn}}$ **return** $\widetilde{\mathbf{F}}_d,\boldsymbol{\mu},\boldsymbol{s}$;

Forward Pass. We implement the UDE module following Algorithm 2. As shown in Fig. 2, the forward pass of the UDE module operates as follows. It takes the depth features $\mathbf{F}_d\in\mathbb{R}^{B\times J\times C_{\text{fuse}}}$ from AMS as input. First, two MLP heads predict the coarse depth $\boldsymbol{\mu}\in\mathbb{R}^{B\times J\times 1}$ and log-variance $\boldsymbol{s}\in\mathbb{R}^{B\times J\times 1}$ for each joint. These scalars are then projected into the feature space and concatenated with $\mathbf{F}_d$ to form $\mathbf{F}_d^{\text{cat}}\in\mathbb{R}^{B\times J\times(3\times C_{\text{fuse}})}$. Finally, a multi-head self-attention block processes this concatenated feature to output the enhanced depth feature $\widetilde{\mathbf{F}}_d\in\mathbb{R}^{B\times J\times C_{\text{fuse}}}$.

3.4. Transformer regressor

Our Transformer-based regression network (Fig. 2) comprises: a Multimodal Interaction Transformer for fusing pose, image, and depth features; a Spatial Transformer for learning global pose representations; and a regression head for 3D coordinate prediction. Following AMS and UDE, all feature sets are unified in dimensionality and spatial representation for subsequent processing. We concatenate $(\mathbf{P}',\{\mathbf{F}_s\}_{s=1}^{S},\widetilde{\mathbf{F}}_d)$ as $\mathbf{F}\in\mathbb{R}^{B\times J\times(S+2)\times C_{fuse}}$.

Multimodal Interaction Transformer. The input feature $\mathbf{F}$ is structured as a $B\times J\times(S+2)\times C_{fuse}$ tensor via stacking, where each joint corresponds to $S+2$ tokens containing tri-modal information. Inspired by multi-modality models that fuse visual and language tokens, we employ a unified transformer encoder to model interactions between 2D pose, depth, and image features. This enables capturing complex cross-modal dependencies and thereby enriches the joint representations. The module implementation proceeds as follows:

$$\mathbf{F}_j'^n=\text{MSA}(\text{LN}(\mathbf{F}_j^{n-1}))+\mathbf{F}_j^{n-1},\tag{10}$$

$$\mathbf{F}_j^n=\text{MLP}(\mathbf{F}_j'^n)+\mathbf{F}_j'^n,\tag{11}$$

where $j(1\ldots J)$ denotes the index of the joint, indicating that feature fusion is performed for each joint. MSA$(\cdot)$ stands for Multi-head Self-Attention. In MSA, input tokens are linearly mapped to Queries $\mathbf{Q}\in\mathbb{R}^{B\times(S+2)\times(C/h)}$, Keys $\mathbf{K}\in\mathbb{R}^{B\times(S+2)\times(C/h)}$, and Values $\mathbf{V}\in\mathbb{R}^{B\times(S+2)\times(C/h)}$, h represents the number of attention heads. Multi-head Attention can

be formulated as:

$$\text{head}_i = \text{softmax}\left(\frac{\mathbf{Q}\cdot\mathbf{K}^{\top}}{\sqrt{d}}\right)\cdot\mathbf{V}, \tag{12}$$

$$\text{MSA}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Concat}(\text{head}_i,\ldots,\text{head}_h). \tag{13}$$

A Spatial Transformer then captures global pose context, reshaping features to $\mathbf{F}\in\mathbb{R}^{B\times J\times(S+2)\cdot C_{fuse}}$ to model skeletal relationships. The final pose is generated by a regression head with linear projection.

3.5. Loss function

Our network is trained using L2 loss to minimize the errors between prediction and the ground truth:

$$\mathcal{L}_{3D} = \frac{1}{J}\sum_{j=1}^{J}\|\hat{y}_j - y_j\|_2, \tag{14}$$

where $\hat{y}_j$ and y_j represent the predicted and ground-truth 3D pose of joint j, respectively, and J is the total number of human body joints. In addition to supervise the uncertainty of the depth features, we introduce Heteroscedastic Uncertainty Loss (HeteroLoss) as $\mathcal{L}_d$. We do not need 'uncertainty labels' to learn uncertainty. Rather, we only need to supervise the learning of the regression task. We learn the log-variance s implicitly from the loss function:

$$\mathcal{L}_d = \frac{1}{J}\sum_{j=1}^{J}\left(\frac{1}{2}\exp(-s_j)\|\mu_j - y_j^{depth}\|^2 + \frac{1}{2}s_j\right), \tag{15}$$

where y_j^{depth} represents the ground-truth depth value of joint j. Specifically, it is defined as the Z-coordinate of the root-relative 3D ground-truth pose and is used in its original metric scale (millimeters) without further normalization. J denotes the total number of joints for a single person. We stabilize the uncertainty learning by (1) scaling 3D coordinate errors by 100, and (2) clipping log-variance s_j to $[-9.0, 9.0]$. The total loss is as $\mathcal{L} = \mathcal{L}_{3D} + \lambda_d\mathcal{L}_d$, where λ_d is weights for $\mathcal{L}_d$.

4. Experiments

4.1. Datasets and evaluation metrics

We evaluate on three standard benchmarks under established protocols. Following previous methods [8,9], on Human3.6M [12], we use subjects S1, S5, S6, S7, S8 for training and S9, S11 for testing, reporting MPJPE (Protocol 1) and PA-MPJPE. On MPI-INF-3DHP [13], we use the indoor and outdoor benchmark with standard train/test split, reporting PCK@150mm, AUC, and MPJPE. On 3DPW [31], we evaluate on the challenging multi-person in-the-wild test set using MPJPE and PA-MPJPE.

4.2. Implementation details

We adopt HRNet-w32 [16] pre-trained on MS COCO [32] as our 2D detector and Depth-Anything V2 [33] for depth estimation. Both models are kept frozen during training without fine-tuning on 3D pose data. In line with previous works [9], we use CPN [5] to obtain the input 2D pose to ensure a fair comparison in the performance evaluation.

Training Configuration. Experiments are conducted on an NVIDIA RTX 4090 GPU using PyTorch. The model is trained for 30 epochs with a batch size of 64 and gradient accumulation over 4 steps (effective batch size 256). We used the AdamW optimizer with an initial learning rate of 6.4×10^{-4}, which is decayed by a factor of 0.99 after each epoch. All grid sampling uses nn.functional.grid_sample with align_corners = True and padding_mode = 'zeros'. For our core modules, the AMS module employs $N_2 = 4$ heads with $N_1 = 5$ sample points per head, a fusion dimension $C_{\text{fuse}} = 128$, and samples from $S = 4$ image feature scales. In the UDE module, the weight for the probabilistic depth loss is set to $\lambda_d = 1\times10^{-5}$.

4.3. Comparison with state-of-the-art methods

Comparison on Human3.6M. Table 2 shows our method achieves SOTA on Human3.6M under Protocol 1 (38.4mm MPJPE) and Protocol 2 (31.4mm PA-MPJPE). To ensure statistical reliability, we report the mean and standard deviation from multiple runs (MPJPE: 38.5 ± 0.14, PA-MPJPE: 31.5 ± 0.09), confirming the consistency of our results. Our approach achieves 20% error reduction (9.6mm absolute improvement) compared to GraphMLP among lifting-based methods, surpassing previous SOTA PSTR by 0.4mm. Against fusion methods, we outperform CA-PF by 3mm (7% relative improvement). For PA-MPJPE, we establish new SOTA, outperforming the latest PSTR baseline by 0.9mm (2.8% relative improvement). These gains validate our depth feature modeling over conventional fusion approaches.

To further investigate the lower bound of our framework, we conduct additional experiments using the ground truth 2D pose as input under the same evaluation protocol 1. In Table 3, our method achieves a new benchmark of 31.2mm MPJPE, surpassing other state-of-the-art approaches in this setting. For instance, it outperforms the recent method PSTR [28], which achieves an MPJPE of 31.5mm, highlighting the effectiveness of our approach.

The cross-metric consistency observed across evaluation protocols (e.g., MPJPE and PA-MPJPE) and input conditions (detected vs. ground-truth 2D poses) demonstrates the stability of our depth feature modeling strategy.

Comparison on MPI-INF-3DHP. Table 4 compares our method with other state-of-the-art approaches on the MPI-INF-3DHP dataset [13], in terms of PCK, AUC, and MPJPE. We use ground truth 2D poses as input, following previous methods [8,9]. Our method achieves the best performance with the MPJPE of 29.4mm, outperforming all compared methods. This includes the state-of-the-art approach relies on 2D pose [28], and surpasses the image features fused method [9]. we further extend the evaluation to include model efficiency metrics. While maintaining moderate complexity (20.4M Params, 0.4G FLOPs), our method reduces MPJPE by 30.3mm compared to FLOPs-heavy PoseFormer [34] using only 29% computational budget. These results demonstrate optimal accuracy without extreme parameterization and strong generalization in challenging environments.

Comparison on 3DPW. As shown in Table 5, our model pretrained on Human3.6M achieves state-of-the-art performance on 3DPW. Our single-frame ASFnet surpasses all compared methods, demonstrating strong generalization. Notably, simply extending to a multi-frame setting (T = 27) with a temporal transformer further improves performance to 108.8 MPJPE and 60.1 PA-MPJPE, underscoring the potential of integrating temporal modeling to enhance robustness in complex in-the-wild scenarios.

4.4. Ablation study

Impact of Each Component. We conduct an ablation study to validate our depth-aware framework and its core components: the Adaptive Multi-Sampling (AMS) module for feature alignment and the Uncertainty-aware Depth Enhancer (UDE) for depth feature refinement.

As shown in Table 6, our Baseline follows prior fusion methods [9], taking 2D pose coordinates and intermediate image features from a 2D pose detector. To align these two modalities, a straightforward single sampling operation is performed on the image feature map at the locations of the 2D joints. These aligned features are then processed through a spatial transformer (e.g., a transformer encoder as in PoseFormer [34]) and an MLP head to estimate the final 3D poses, achieving an MPJPE of 41.5mm. Incorporating depth features via a dedicated estimator with basic sampling and concatenation reduces MPJPE by 1.6mm. Integrating AMS further improves performance to 39.4mm by enhancing feature extraction effectiveness. Notably, adding UDE alone yields limited improvement (40.2mm), revealing a key dependency: UDE's uncertainty modeling requires the rich, accurately-located features pro-

Table 2
Results on Human3.6M dataset in millimeters under Protocol 1&2. 2D pose detected by CPN [5] are used as input. Top-table: results under Protocol 1 (MPJPE). Bottom-table: results under Protocol 2 (PA-MPJPE). The best results are highlighted in bold, and the second-best are underlined. Ours achieves the best performance.

Protocol 1	Venue	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg
Anatomy [18]	TCSVT'21	41.4	43.5	40.1	42.9	46.6	51.9	41.7	42.3	53.9	60.2	45.4	41.7	46.0	31.5	32.7	44.1
PoseFormer [34]	ICCV'21	41.5	44.8	39.8	42.5	46.5	51.6	42.1	42.0	53.3	60.7	45.5	43.3	46.1	31.8	32.2	44.3
Geometric [35]	TCSVT'23	45.3	51.8	46.4	49.9	51.4	57.6	49.1	46.5	59.1	67.9	51.0	48.9	53.0	38.0	40.6	50.4
CA-PF [9]	NeurIPS'23	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	41.4
DGFormer [7]	PR'24	46.3	50.3	45.7	50.5	50.8	57.5	49.6	46.0	55.8	63.8	50.9	47.8	53.0	38.7	41.3	49.8
GraphMLP [8]	PR'24	43.7	49.3	45.5	47.9	50.5	56.0	46.3	44.1	55.9	59.0	48.4	45.7	51.2	37.1	39.1	48.0
PSTR [28]	TCSVT'25	36.3	38.7	38.3	**31.5**	**39.4**	51.1	37.1	35.4	51.8	57.4	**39.6**	**36.3**	**24.7**	39.1	**25.4**	38.8
ASFnet (Ours)	–	**34.0**	**37.5**	**34.9**	38.5	39.7	**44.2**	**33.6**	**35.3**	**45.6**	**51.5**	40.0	37.6	39.8	**30.6**	32.3	**38.4**
Protocol 2	Venue	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg
Anatomy [18]	TCSVT'21	32.6	35.1	32.8	35.4	36.3	40.4	32.4	32.3	42.7	49.0	36.8	32.4	36.0	24.9	26.5	35.0
PoseFormer [34]	ICCV'21	32.5	34.8	32.6	34.6	35.3	39.5	32.1	32.0	42.8	48.5	34.8	32.4	35.3	**24.5**	26.0	34.6
Geometric [35]	TCSVT'23	35.4	38.8	35.9	40.4	39.6	44.0	36.7	35.4	46.8	53.7	40.9	36.6	42.0	30.6	33.9	39.4
CA-PF [9]	NeurIPS'23	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	33.5
DGFormer [7]	PR'24	35.4	38.4	35.8	40.3	39.2	43.7	37.6	34.8	44.7	51.3	40.2	36.1	41.2	30.6	33.9	38.9
GraphMLP [8]	PR'24	35.1	38.2	36.5	39.8	39.8	43.5	35.7	34.0	45.6	47.6	39.8	35.1	41.1	30.0	33.4	38.4
PSTR [28]	TCSVT'25	29.2	31.9	31.3	**26.4**	36.6	37.7	28.8	29.6	40.3	47.3	37.6	**29.4**	**24.1**	28.9	**24.7**	32.3
ASFnet (Ours)	–	**28.9**	**31.0**	**29.6**	31.5	**32.0**	**34.8**	**27.5**	**28.5**	**38.4**	**41.5**	**33.2**	29.5	32.6	**24.5**	27.1	**31.4**

Table 3
Results on Human3.6M dataset in millimeters under Protocol 1 (MPJPE). Ground truth 2D pose are used as input. The best results are highlighted in bold, and the second-best are underlined.

GT Protocol 1	Venue	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg
Anatomy [18]	TCSVT'21	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	32.3
PoseFormer [34]	ICCV'21	30.0	33.6	29.9	**31.0**	30.2	33.3	34.8	31.4	37.8	38.6	31.7	**31.5**	**29.0**	**23.3**	**23.1**	31.3
Geometric [35]	TCSVT'23	30.0	35.5	30.1	31.9	34.8	37.3	35.4	31.7	39.5	42.9	33.3	34.4	34.6	28.5	29.9	34.0
CA-PF [9]	NeurIPS'23	34.7	31.8	**25.4**	36.9	31.1	**32.0**	31.6	**27.7**	**29.0**	39.8	30.2	36.8	31.5	27.2	28.9	31.7
DGFormer [7]	PR'24	31.5	34.3	28.2	32.2	31.3	36.8	37.0	29.4	34.9	37.8	31.8	32.5	33.0	26.7	28.9	32.4
GraphMLP [8]	PR'24	32.2	38.2	29.3	33.4	33.5	38.1	38.2	31.7	37.3	**38.5**	34.2	36.1	35.5	28.0	29.3	34.2
PSTR [28]	TCSVT'25	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	31.5
ASFnet (Ours)	–	**29.7**	**30.1**	26.0	37.1	**29.2**	33.3	**28.9**	28.6	30.6	47.0	**29.1**	35.3	30.2	25.2	26.7	**31.2**

Table 4
Comparison on MPI-INF-3DHP dataset with 2D ground truth inputs. PCK, AUC, and MPJPE are used as main evaluation metrics, with model complexity (Params/M, FLOPs/G) reported for reference. Best in bold, second-best underlined.

Method	Venue	Parameters(M)	FLOPs (G) for per Frame	PCK (↑)	AUC (↑)	MPJPE (↓)
Anatomy [18]	TCSVT'21	59.2	0.1	87.8	53.8	79.1
PoseFormer [34]	ICCV'21	9.6	1.4	94.3	62.7	59.7
Geometric [35]	TCSVT'23	2.4	0.04	96.7	75.6	38.2
CA-PF [9]	NeurIPS'23	14.1	0.3	98.0	75.4	32.7
DGFormer [7]	PR'24	4.3	-	85.5	53.6	80.4
GraphMLP [8]	PR'24	-	0.4	87.0	54.3	–
PSTR [28]	TCSVT'25	35.1	-	**98.9**	**77.5**	30.1
ASFnet (Ours)	–	20.4	0.4	98.3	77.1	**29.4**

Table 5
Evaluation on the in-the-wild 3DPW dataset. Results are reported in MPJPE and PA-MPJPE (mm) using models trained on Human3.6M without fine-tuning. T denotes the number of input frames. Best results are in bold; second-best are underlined.

Method	Venue	T	MPJPE (↓)	PA-MPJPE (↓)
PoseFormer [34]	ICCV'21	81	118.2	73.1
CA-PF [9]	NeurIPS'23	1	127.5	72.6
DGFormer [7]	PR'24	1	154.9	77.6
GraphMLP [8]	PR'24	1	140.5	66.7
ASFnet (Ours)	–	1	116.4	62.9
ASFnet (Ours)	–	27	**108.8**	**60.1**

Table 6
Ablation study on key components of our depth-aware framework. Baseline refers to previous methods that solely rely on a 2D pose detector to generate 2D poses and image features. We ablate the following key components of our framework: AMS (Adaptive Multi-Sampling), and UDE (Uncertainty-aware Depth Enhancer). MPJPE is reported in millimeters.

Configuration	Depth Features	AMS	UDE	MPJPE (↓)
Baseline (2D pose + image features)	×	×	×	41.5
+ Depth features	✓	×	×	39.9 (↓ 1.6)
+ Depth features + AMS	✓	✓	×	39.4 (↓ 2.1)
+ Depth features + UDE	✓	×	✓	40.2 (↓ 1.3)
Full model (Ours)	✓	✓	✓	**38.4** (↓ 3.1)

vided by AMS. With single-sampled features, UDE lacks sufficient input diversity and spatial precision, constraining its performance gain. Combining both modules achieves our best result (38.4mm MPJPE), where UDE's noise suppression and adaptive fusion of AMS-refined features delivers a total 3.1mm improvement.

Impact of Depth Estimator Precision on Model Performance. Fig. 4 demonstrates that our method ensures stable effectiveness across

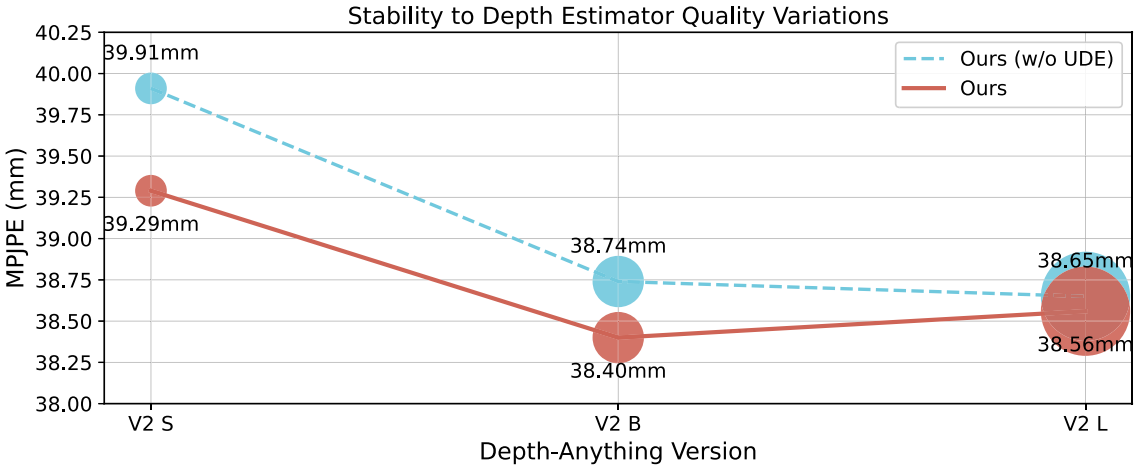


Fig. 4. Impact of depth estimator precision on 3D pose estimation accuracy (MPJPE in mm). Solid and dashed curves represent our full model and the baseline without UDE, respectively, both evaluated with three depth estimator variants (the Small, Base, and Large versions of Depth-Anything V2 [36]). Marker sizes correspond to depth estimator parameter counts. Our UDE module consistently improves performance across all estimators (e.g., V2S: 39.91→39.29; V2B: 38.74→38.40; V2L: 38.65→38.56). It demonstrates superior adaptability to depth estimator quality variations.

Table 7
Comparison of inference time (measured on single NVIDIA RTX 4090 GPU) and MPJPE for our method with depth-independent methods. The additional overhead from integrating depth estimation is acceptable while providing significant performance improvements.

Model	Stage 1: 2D + Depth Estimation (ms)			Stage 2: Lifting (ms)	Total inference Time (ms)	MPJPE (↓)
	2D Pose Est.	Depth Est.	Subtotal			
CA-PF [9]	29.6	–	29.6	5.2	33.1	41.4
ASFnet (Ours) *with Depth-Anything V2-Small*	29.6	10.9	36.5	5.9	43.1	39.3
ASFnet (Ours) *with Depth-Anything V2-Base*	29.6	23.8	36.9	5.9	43.2	**38.4**

depth estimators of varying accuracies. A key observation is that the UDE module consistently enhances performance regardless of the estimator's precision, reducing the MPJPE for all three variants (V2S: 39.91→39.29; V2B: 38.74→38.40; V2L: 38.65→38.56). This shows that UDE effectively mitigates performance degradation from imperfect depth estimation. With the smallest depth estimator (Depth-Anything V2-Small, 39.3mm MPJPE), our full model achieves 5.3% lower error than the no-depth baseline (41.5mm), proving effective depth utilization even with limited estimator capacity. Furthermore, UDE reduces the performance gap between high- and low-precision depth estimators, enhancing pipeline robustness and enabling lighter estimators to be deployed with only marginal accuracy loss.

Table 7 compares inference speed on an NVIDIA RTX 4090 GPU. Specifically, integrating the Depth-Anything V2-Base depth estimator introduces an additional 7.3 ms per frame in Stage 1, representing a 24.7% increase in time over the depth-independent baseline. Nevertheless, even with our depth-aware modules (AMS and UDE) in Stage 2, the total inference time rises by only 10.1 ms (from 33.1 ms to 43.2 ms), which demonstrates that the computational overhead introduced by depth estimation remains manageable and does not become a system-wide bottleneck. Moreover, our sparse multi-sampling (AMS) and Uncertainty Regression (UDE) introduce insignificant additional cost in the lifting stage (Stage 2), as reflected in the comparable lifting times (5.2 ms vs. 5.9 ms) and corroborated by the FLOPs/parameters compared with other mehtods reported in Table 4. Notably, we achieve optimal accuracy (38.4mm MPJPE) with mid-sized Depth-Anything V2-Base, achieving a relative MPJPE improvement of 7.2% over the depth-independent baseline, demonstrating that UDE's feature refinement compensates for imperfect depth priors. This benefits resource-constrained deployments preferring lightweight estimators, confirming our architecture converts depth errors into learnable features rather than propagation pathways, as further validated in Fig. 10.

Table 8
Ablation study on AMS module design: Comparison with alternative attention and sampling strategies. The proposed Adaptive Multi-Sampling significantly outperforms both adaptive feature selection and attention mechanisms baselines by mitigating background noise and robustness to 2D pose errors.

Module Variant	MPJPE (↓)
Adaptive feature selection	49.8
Cross-attention	45.3
Single sampling + Self-attention	44.2
Single sampling only	40.2
AMS (Ours)	**38.4**

Ablation Study on AMS. We evaluate our Adaptive Multi-Sampling (AMS) module against conventional approaches in Table 8. Four baselines are compared: (1) Adaptive feature selection from global depth maps (49.8mm MPJPE), suffering from background noise; (2) Cross-attention with 2D pose queries (45.3mm), similarly noise-sensitive; (3) Single sampling + self-attention (44.2mm), reducing noise but vulnerable to 2D pose errors; (4) Plain single sampling (40.2mm), improved but still error-sensitive. Our AMS achieves superior performance (38.4mm) via iterative corrective sampling that dynamically refines features while maintaining robustness to imperfect 2D inputs, demonstrating substantial gains over both attention-based and sampling-based methods.

Resilience to Controlled Depth Noise. We evaluate robustness against realistic depth imperfections: motion blur and pixel loss. To simulate motion artifacts, a Gaussian blur is applied to random regions (20% image width) with varying kernels (3×3, 5×5, 7×7). Table 9 shows stable MPJPE (38.4–38.5mm) and marginal MPJVE

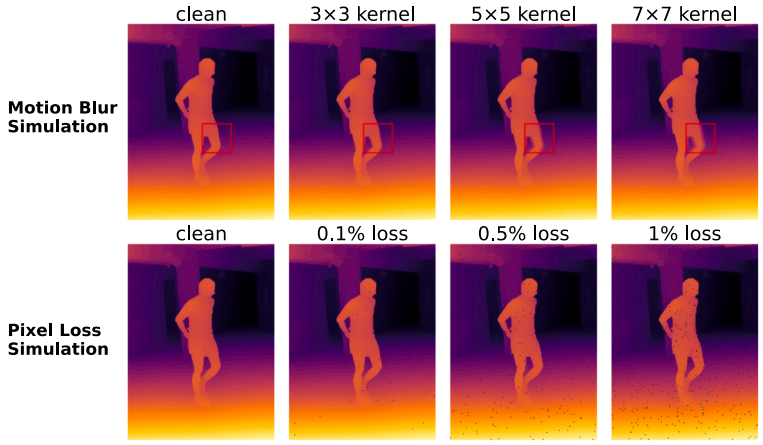


Fig. 5. Visual examples of synthetic noise applied to depth maps for robustness evaluation. *(Top row)* Progressive motion blur using 3×3, 5×5, and 7×7 Gaussian kernels. *(Bottom row)* Progressive random pixel loss at rates of 0.1%, 0.5%, and 1%.

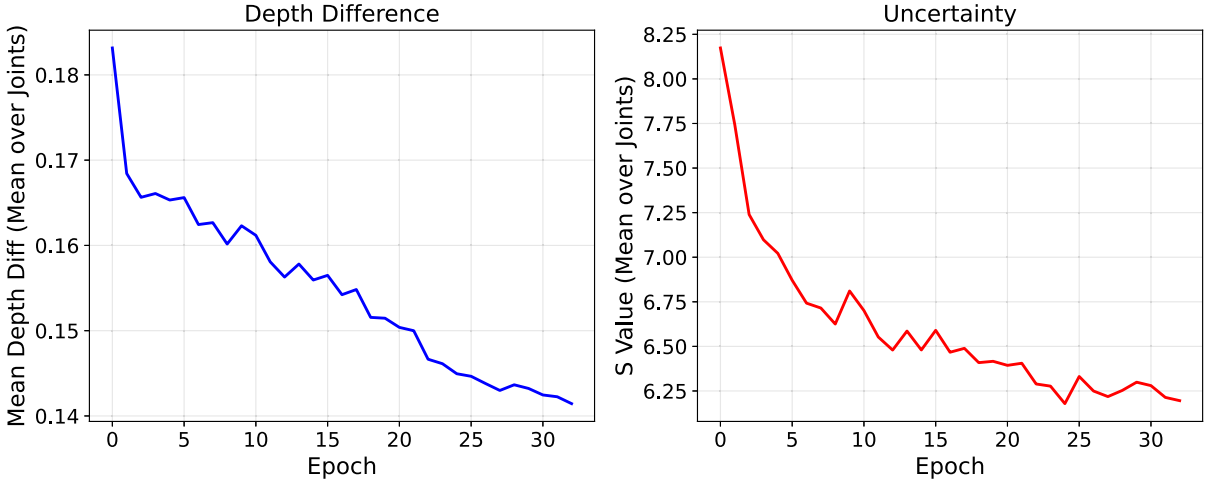


Fig. 6. Training dynamics showing stable co-evolution of prediction error and uncertainty. Both metrics decrease smoothly and remain bounded, demonstrating effective uncertainty learning without divergence or collapse.

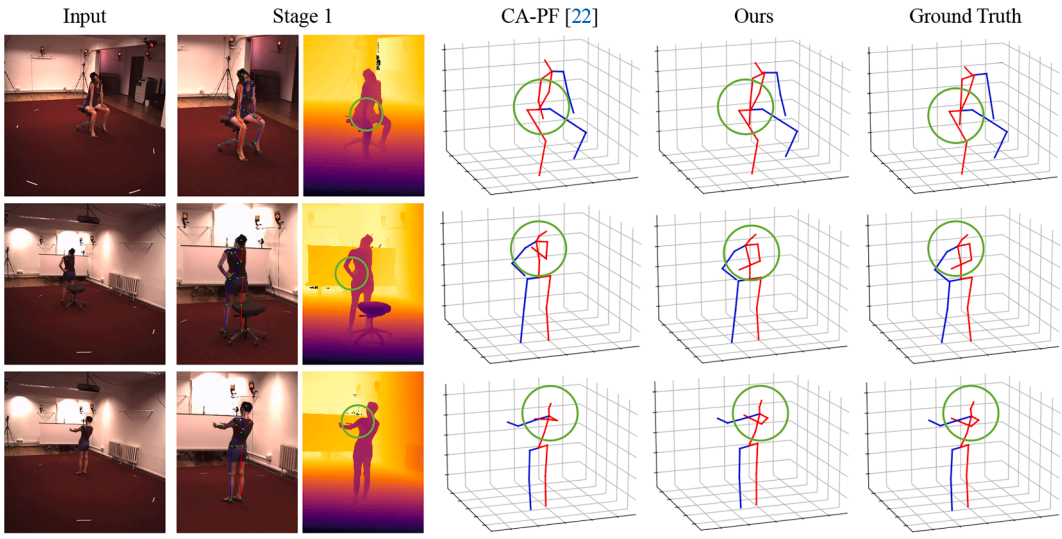


Fig. 7. Qualitative comparison between our method and the state-of-the-art fusion method CA-PF [9] on Human3.6M dataset. Green circles highlight key improvements: (a) Our method alleviates depth ambiguity in the right arm (Row 1), correctly inferring its extended 3D position rather than erroneous bending; (b) Precise localization of the occluded right wrist at the waist through depth gradient analysis (Row 2); (c) Recovery of forearm position from depth discontinuity between right arm and torso (Row 3). Explicit depth feature modeling enables physically plausible estimations where planar features fail.

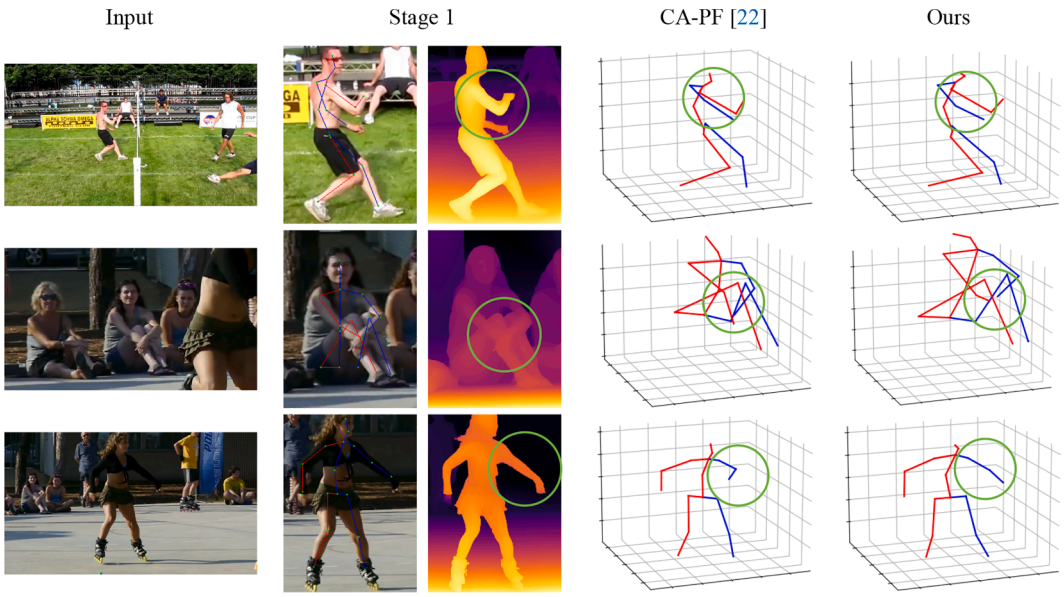


Fig. 8. Qualitative comparison on noisy in-the-wild data between our method and the state-of-the-art fusion method CA-PF [9]. Green circles highlight key advantages: (a) reliable depth ordering of left/right arms (Row 1); (b) precise differentiation of elbow-body depth relationships under complex self-occlusion scenarios (Row 2); (c) consistent depth perception in cluttered scenes (Row 3).

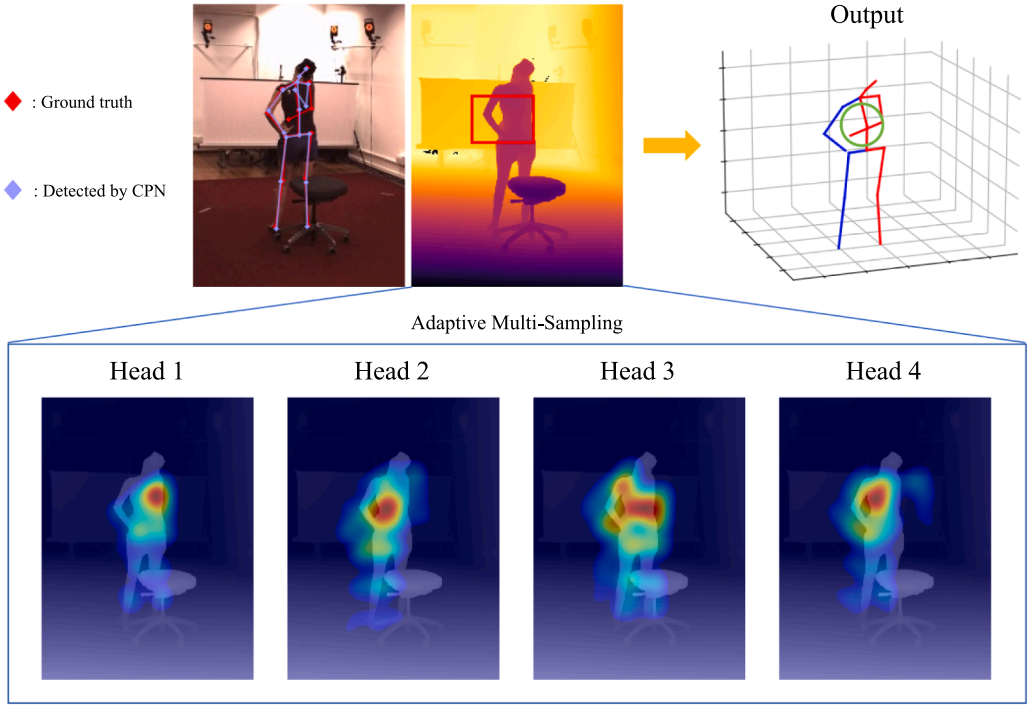


Fig. 9. Visual analysis of the AMS module. Visualization of sampling points learned by each head in the Adaptive Multi-Sampling (AMS) module. The heatmaps on the depth map in the blue box show the learned sampling points, where brighter regions indicate higher weights. Each head learns to focus on different regions relevant to the human pose. As illustrated, although the input 2D pose contains incorrect wrist information, head 2 learns to attend to the true wrist joint region, effectively suppressing noise from the erroneous 2D detection. This enables the extraction of reliable depth features and contributes to more accurate 3D pose estimation. The results demonstrate that AMS can dynamically adjust sampling behavior, focusing on informative regions, not limited by the estimated 2D pose. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

increase (7.43-7.83mm), confirming motion distortion invariance. Examples of these simulated perturbations are shown in Fig. 5 (top row). To evaluate robustness against corrupted depth data, we randomly zero 0.1-1% of depth pixels. Fig. 5 (bottom row) illustrates examples of the pixel loss patterns. Table 10 shows maintained MPJPE at 1‰ loss (38.4mm) and controlled degradation at 1% (38.7mm), with predictable MPJVE progression (7.43-9.67mm).

Our evaluation establishes the method's dual strength: while achieving optimal performance with precise depth estimators, it maintains remarkable robustness under practical noise conditions through

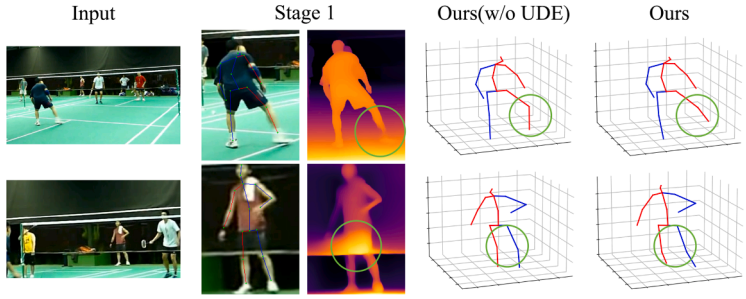


Fig. 10. Visual analysis of the UDE module. Qualitative comparison between our full method, which incorporates Uncertainty-aware Depth Enhancer (UDE), and its ablated version (without UDE) in challenging depth-error scenarios of in-the-wild data. Green circles highlight critical improvements enabled by UDE: (a) Recovery of the right ankle under fast motion (Row 1), where missing depth data misleads the baseline into predicting an unnatural vertical right calf, while UDE avoids the interference of erroneous depth information; (b) Correction of knee bending artifacts caused by depth discontinuities from net-edge occlusion (Row 2), where UDE filters corrupted depth cues to restore biomechanically valid pose.

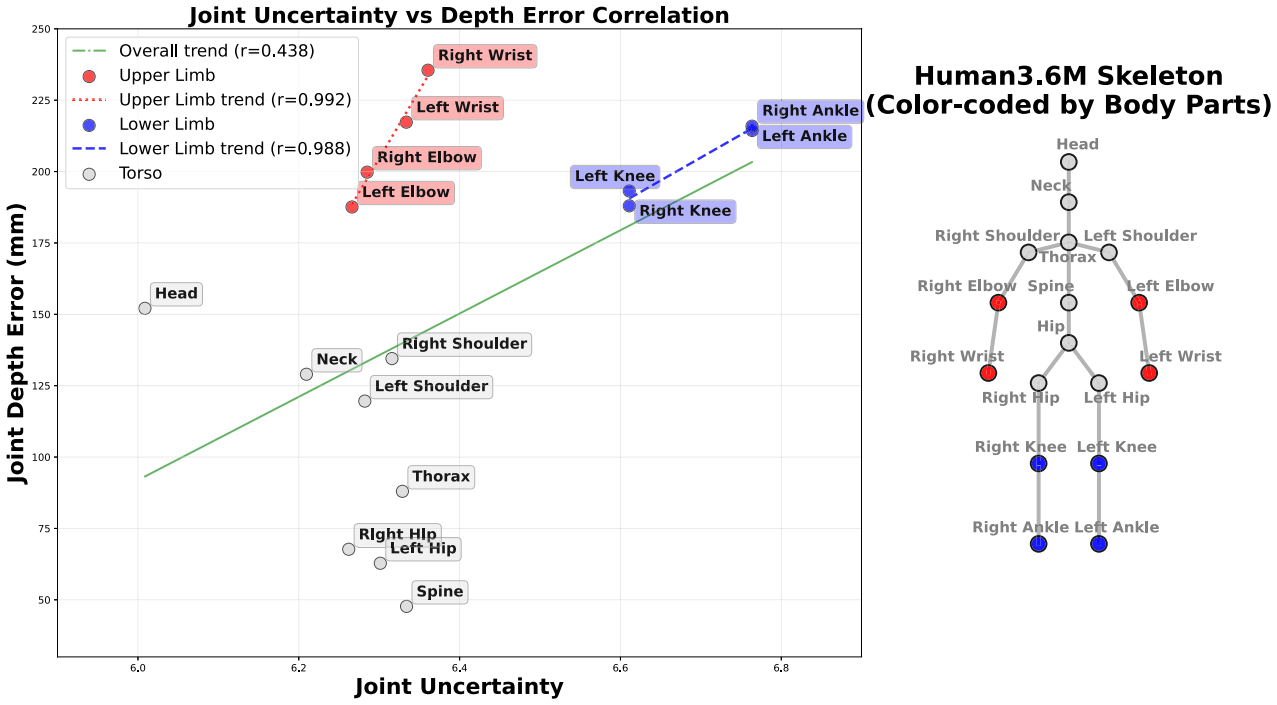


Fig. 11. Correlation analysis between mean joint uncertainty (s) and mean coarse joint depth (μ) error on the Human3.6M validation set. (Left) Scatter plot showing the correlation of joint uncertainty and joint depth error across all joints. (Right) Human3.6M skeleton colored by body parts. The observed positive correlation, particularly in high-error limb joints, validates that our learned uncertainty reflects depth feature reliability.

Table 9
Robustness analysis under motion blur noise. Our method maintains stable MPJPE across all blur levels.

Blur Kernel Size	MPJPE (↓)	MPJVE (↓)
No noise	38.4	7.43
3 × 3	38.4	7.46
5 × 5	38.4	7.57
7 × 7	38.5	7.83

Table 10
Robustness analysis under missing data noise. The method demonstrates remarkable resilience to data loss while maintaining MPJPE.

Missing Probability	MPJPE (↓)	MPJVE (↓)
No noise	38.4	7.43
0.1%	38.4	7.68
0.5%	38.6	8.62
1%	38.7	9.67

uncertainty-aware depth enhancement, ensuring reliable real-world applicability.

Training Stability and Uncertainty Analysis. To address the stability of our uncertainty estimation during training, we monitor the log-variance s_j throughout optimization. Fig. 6 shows that both depth prediction error and uncertainty estimates exhibit stable co-evolution across epochs. The mean depth difference consistently decreases, indicating improved pose accuracy, while uncertainty estimates decrease in a tightly coupled manner. This parallel decline confirms that our uncertainty effectively tracks actual error as a meaningful confidence measure. Crucially, the log-variance s remains within a stable and bounded range throughout the entire training.

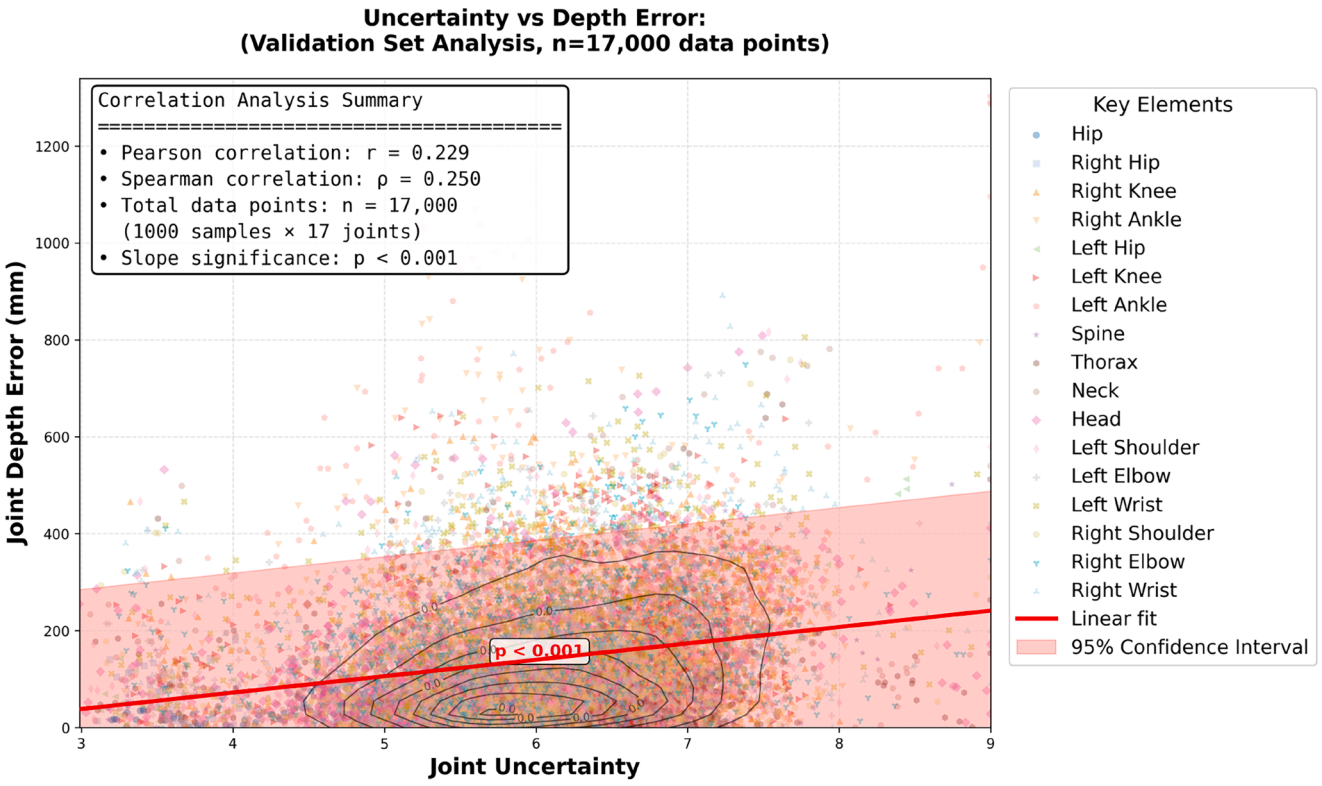


Fig. 12. Correlation between joint uncertainty and depth error at the individual-joint level on the Human3.6M validation set. Scatter plot showing the correlation for 17,000 individual joints (sampled from 1000 poses). Statistically significant positive correlation ($r = 0.229$, $p < 0.001$) is observed, with dense concentration in the lower-triangular region, confirming that higher uncertainty indicates larger errors.

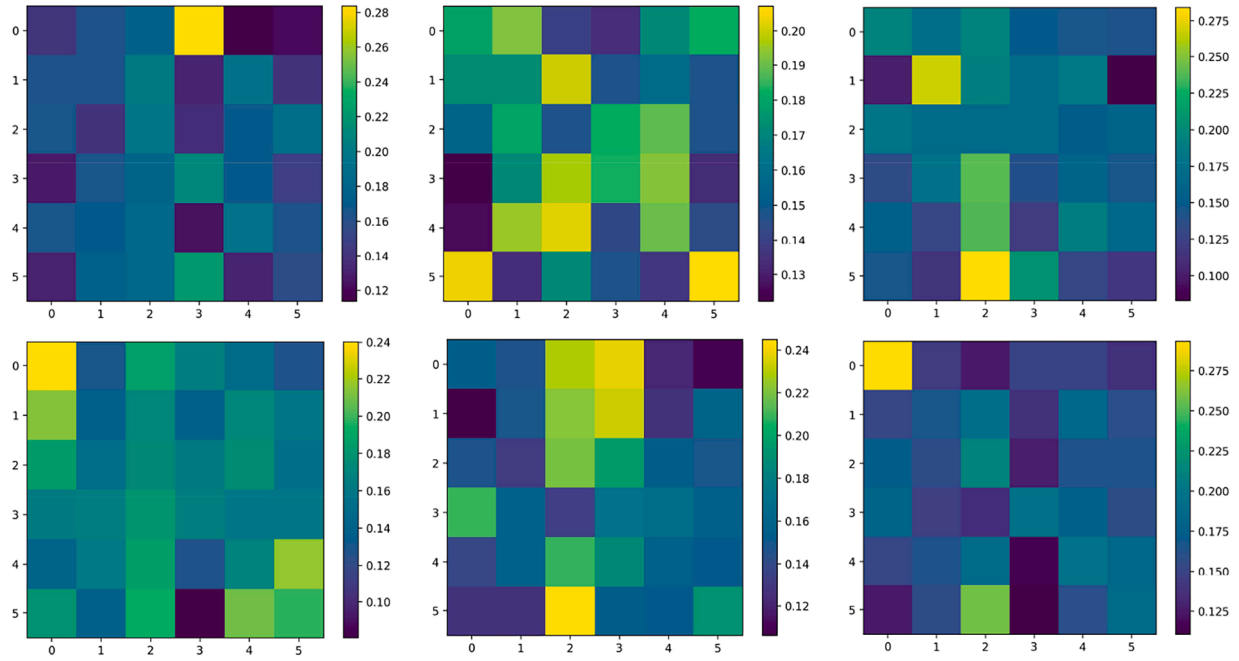


Fig. 13. Visualization of the attention maps from the Multimodal Interaction Transformer in our model. It illustrates that the self-attention mechanism systematically assigns a weight distribution to multimodal features for each joint, leading to exploit complementary multimodal cues, all of which contribute to the inference. We randomly select six joints as an example; brighter colors indicate higher attention scores.

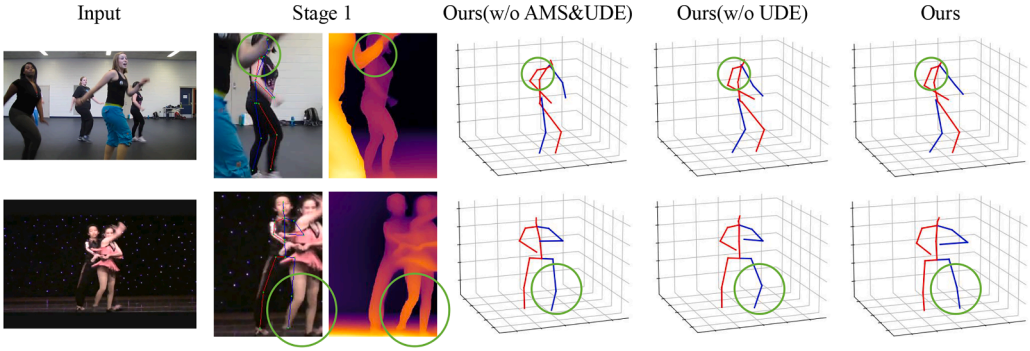


Fig. 14. Qualitative results under multi-person occlusion. Top: AMS corrects 2D pose errors from arm occlusion, resolving the hunched posture in "w/o AMS&UDE." Bottom: UDE mitigates depth distortion from leg occlusion, improving leg pose in "Ours" over "w/o UDE." This case also reveals the inherent limitation of single-frame methods under severe occlusion, where visual cues (both 2D pose and depth) are nearly missing for the occluded limb.

4.5. Qualitative analysis

Comparisions on Human3.6M dataset. Fig. 7 presents qualitative comparisons on Human3.6M with the fusion-based CA-PF [9]. Our method produces more realistic 3D poses by leveraging depth-aware representations, which successfully resolve depth ambiguities and capture subtle anatomical details that are missed by purely planar fusion methods.

Specifically, In Row 1, we correctly reconstruct the arm's extended 3D configuration, while CA-PF bends it erroneously. In Row 2, even with incorrect 2D inputs, our method locates the wrist accurately by interpreting the depth gradient at the left waist (green circle)-a distinction planar fusion misses, causing CA-PF to place it near the shoulder. Row 3 further shows that sharp depth discontinuities between the right arm and torso enable accurate forearm reconstruction under occlusion, whereas CA-PF yields an implausible elbow. By modeling robust depth features, our method resolves 3D relationships that remain ambiguous in 2D cues and planar fusion.

Comparisons on Wild Images. Fig. 8 validates our method's generalization on in-the-wild images versus CA-PF [9]. ow 1 demonstrates our method's ability to resolve depth ambiguity in overlapping arms, accurately estimating their relative distances-a task CA-PF fails at. In Row 2, our model leverages the depth discontinuity at the occluded elbow (green circle) to infer a plausible pose, while CA-PF's limited planar reasoning causes it to fuse the wrist with the torso. Row 3 confirms robustness in cluttered scenes, where our method maintains consistent depth perception. By integrating depth features with 2D poses, our approach overcomes the limitations of pure planar fusion, ensuring effectiveness across diverse cases.

Visual analysis of Adaptive Multi-Sampling. Fig. 9 visualizes the sampling points learned by our Adaptive Multi-Sampling (AMS) module. The heatmap in the blue box provides an overview, showing that each head adaptively learns distinct sampling strategies from the depth map. This enables the module to flexibly focus on pose-relevant regions, effectively compensating for input noise (as evidenced by the inputs and result in the blue box) and learning more comprehensive, context-aware features.

Visual analysis of Uncertainty-aware Depth Enhancer. Fig. 10 validates our UDE against an ablated variant. In Row 1, UDE suppresses unreliable depth features at a motion-missed ankle to predict a natural pose, while the baseline incorrectly places it vertically downward. In Row 2, UDE corrects a false knee bend induced by occlusion, recovering the leg's true configuration. This confirms that our uncertainty-based feature enhancement is key to robust performance in challenging in-the-wild scenarios.

To further quantitatively validate that our learned uncertainty reflects estimation reliability, we conducted a comprehensive correlation analysis on the Human3.6M validation set between predicted uncertainty and actual depth errors. As shown in Fig. 11 (per-joint-type averages), we observe a clear positive correlation between joint uncertainty and depth estimation error across all joints. The results reveal distinctive regional patterns: body parts with typically larger depth errors-namely the upper and lower limbs-exhibit remarkably strong positive correlations ($r = 0.992$ and $r = 0.988$, respectively). This confirms that our uncertainty metric faithfully represents the reliability of joint depth features, particularly in challenging limb regions. Furthermore, the large-scale scatter plot in Fig. 12 (17,000 individual joints) statistically reinforces this overall positive trend ($r = 0.229$, $p < 0.001$). The dense concentration of points in the lower-triangular region provides direct visual and statistical evidence that higher predicted uncertainty consistently corresponds to larger depth errors.

Table 11
Performance improvement by integrating our (AMS + UDE) modules into different lifting architectures on Human3.6M dataset. *T* denotes the number of input frames. MPJPE and PA-MPJPE are reported in millimeters.

Method	T	MPJPE(↓)	PA-MPJPE (↓)
GraphMLP [8], PR'24	1	48.0	38.4
+ (AMS + UDE) (**Ours**)	1	**39.1**	**32.4**
MotionAGFormer-XS [37], WACV'24	27	45.1	36.9
+ (AMS + UDE) (**Ours**)	27	**42.6**	**35.3**

Visual analysis of Multimodal Interaction Transformer. We visualized the attention maps from the Multimodal Interaction Transformer of our proposed model. Each attention map extracted at every joint is a 6×6 matrix, representing the weight distribution assigned by the self-attention mechanism to the multimodal features. In Fig. 13, each joint focuses on different feature maps, and the Multimodal Interaction Transformer is capable of capturing diverse features from the multimodal inputs for each joint.

4.6. Modularity and generalizability analysis

We evaluate the modularity of our AMS and UDE modules by integrating them into multiple state-of-the-art frameworks, demonstrating their effectiveness as plug-and-play components that improve performance across architectures. As shown in Table 11, our modules bring consistent gains to diverse backbones. For single-frame GraphMLP, MPJPE drops 8.9mm to 39.1mm. for multi-frame MotionAGFormer-XS (27 frames), it improves from 45.1mm to 42.6mm. Notably, these improvements are achieved while AMS and UDE maintain a purely frame-wise processing logic, without any temporal extension. This demonstrates our generalizability in effectively mitigating fundamental issues of 2D pose noise and depth ambiguity.

Table 12
Multi-frame 3D pose estimation results on Human3.6M under protocol 1. 2D poses are detected by CPN. 'T' denotes the input sequence length. Best results are in bold, second-best are underlined.

Method	Venue	MPJPE (↓)
PoseFormer [34] (T = 81)	ICCV'21	44.3
GraphMLP [8] (T = 243)	PR'24	43.8
STGFormer [38] (T = 243)	PR'25	40.3
PSTR [28] (T = 243)	TCSVT'25	38.8
Spectral [39] (T = 243)	PR'26	<u>37.7</u>
ASFnet (Ours) (T = 1)	–	38.4
ASFnet (Ours) (T = 27)	–	**37.6**

4.7. Analysis in multi-person occlusion scenarios

To analyze the role of AMS and UDE in complex multi-person scenarios and clarify the limitations of our method, we conduct a qualitative study on two in-the-wild occlusion cases (Fig. 14). In the moderate occlusion case (first row), Arm occlusion causes erroneous 2D thorax-spine pose. Without AMS (Ours w/o AMS&UDE), the model yields a hunched pose. AMS corrects it by multi-sampling to distinguish the depth difference between the thorax and the occluding arm, recovering natural pose (Ours w/o UDE and Ours). In the severe occlusion case (second row), the left leg of the rear person is heavily occluded, making both 2D pose and depth features unreliable. AMS alone fails due to distorted depth; UDE down-weights unreliable depth and integrates context, yielding plausible pose (Ours)-though the precise extension of the leg remains inaccurate. Even with both modules, severe occlusion prevents precise limb recovery-single-frame spatial context encounters an information bottleneck when 2D and depth cues are nearly absent.

4.8. Temporal extension of the proposed method

To demonstrate extensibility, we simply adapt our framework to multi-frame input by adding a standard Temporal Transformer (commonly used in multi-frame methods like PoseFormer [34]) after the Spatial Transformer, using $T = 27$ frames. Crucially, both the AMS and UDE modules remain single-frame based without any temporal extension. As shown in Table 12, this extension achieves 37.6mm MPJPE, outperforming our single-frame baseline and previous state-of-the-art methods-even those using longer sequences ($T = 243$). This validates our framework's compatibility with temporal modeling, forming a competitive multi-frame pipeline.

Future Work Our multi-frame extension confirms the temporal compatibility of our framework, paving the way for deeper integration. Future work will develop a Temporal-Adaptive Multi-Sampling (T-AMS) module to sample spatiotemporally coherent features from joint trajectories, and a Temporal Uncertainty-aware Depth Enhancement (T-UDE) module to fuse uncertainty estimates across frames. These advancements are expected to deliver significant gains in accuracy and temporal stability.

5. Conclusion

This paper presents ASFnet, a novel depth-aware architecture for monocular RGB-based 3D human pose estimation that explicitly models robust depth features to mitigate depth ambiguity. By introducing the Adaptive Multi-Sampling (AMS) module, our approach effectively extracts pose-relevant depth information while reducing dependence on noisy 2D pose estimates. Additionally, the Uncertainty-aware Depth Enhancer (UDE) further refines depth features by adaptively estimating and suppressing uncertainty-driven noise. The fusion of these enhanced depth cues with 2D pose information via a Transformer regressor enables stable 3D pose estimation. Extensive evaluations on the Human3.6M and MPI-INF-3DHP benchmarks demonstrate that ASFnet achieves competitive performance compared to state-of-the-art methods, highlighting the potential of explicit depth modeling for advancing monocular 3D human pose estimation.

Despite the effectiveness of our approach, we acknowledge its limitations. Although our method effectively alleviates depth ambiguity in monocular 3D human pose estimation, its current single-frame design inherently limits its capability to handle severe inter-person occlusions commonly found in multi-person scenarios.

Looking forward, this limitation reveals a promising direction. Temporal information across consecutive frames provides crucial motion cues that help resolve challenging cases and stabilize pose trajectories. Building on the encouraging results of our initial multi-frame extension, future work will focus on developing temporally enhanced versions of our core modules (e.g., T-AMS and T-UDE) to explicitly model human motion continuity. We believe this will lead to more robust pose estimation in complex, in-the-wild environments, and we hope our work inspires further research in this direction.

CRediT authorship contribution statement

Mengyuan Liu: Writing – original draft, Visualization, Project administration, Methodology, Formal analysis; **Jingting Liu:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis.

Data availability

I have shared the link to my code at the manuscript

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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